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Temperature transients in TRISO type fuel

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ABSTRACT

The High Temperature Gas-cooled Reactor (HTGR), cooled by gas and moderated by graphite, is the predecessor of the Generation IV Very High Temperature Reactor (VHTR). Recently, the South African project of the Pebble-Bed Modular Reactor (PBMR) seems to merge in high temperature, helium cooled and micro-fuel particles called TRISO in pebble-bed. TRISO particles must not exceed 2000 °C due to safety requirements. The inner structure of the fuel element is sufficiently complex to know with a certain uncertainty its heat transfer coefficients. It also explains that the in situ heat transfer time can differ from the one obtained in the laboratory. Hence, a mesoscale simplified analytical model (pebble scale) has been developed in this work in order to study temperature transients in operation. Its assigned coefficients have a certain experimental support. The trial function with physical meaning method is used to solve the model equations and the in situ estimation of the coefficients is achieved by means of thermal noise analysis from thermocouple signals. Instead of complex numerical models, the approximate solutions are very simple and enough for safety requirements and surveillance. Both heat transfer time and conductivity are estimated so that the burnup level of a certain element can be monitored. A case study is presented using data obtained from the literature. This result permits selecting the sampling time of the noise signals, and also optimizing an Autoregressive Moving Average (ARMA) (1,1) model according to the solution provided by the simplified model.

1. Introduction

The Very High Temperature Reactor (VHTR) is one of the six Generation IV proposed concepts (US DOE, 2002). The HTGR (High Temperature Gas-cooled Reactor) was the predecessor, cooled by gas and moderated by graphite like the AVR in Germany, of 49 MWth, shut down after Chernobyl accident. Recently, gas cooled reactors have gained interest and certain designs like the Pebble-Bed Modular Reactor (PBMR) have been developed. The South African PBMR project has just terminated in September 2010 after 12 years of development (Thomas, 2011).

The basic features of VHTRs are helium cooled, graphite moderated with a ceramic core and TRISO coated fuel particles. The outlet temperatures reach up to 1000 °C which make these reactors attractive for secondary applications such as hydrogen manufacturing, actinide transmutation and besides they are not proliferant (Garcia Fajardo, 2012; Abánades and Pérez-Navarro, 2007).

The triple-isotropic (TRISO) coated fuel particles consist of a uranium fuel kernel surrounded by three layers of pyrolytic carbon. They are dispersed in a spherical graphite matrix to form the fuel elements, generally the size of a tennis ball. They must not exceed 2000 °C due to safety requirements since there are a number of known fuel failure and fission product release mechanisms that are temperature and burnup dependent (Powers and Wirth, 2010; Maki et al., 2007); therefore, all the conceptual designs which use this fuel must carry out heat transfer studies. Some of these studies can be found in Ortensi et al. (2011), Ortensi and Ougouag (2009), Stainsby et al. (2008), Stainsby et al. (2010) and they model the temperature profiles over a wide range of length scales; micro (particle), meso (pebble) and macro (several pebbles) scale.

As the inner structure of the fuel element is rather complex, there is a certain uncertainty in knowing the heat transfer coefficients. Hence, complex models of numerical calculus are not necessarily better than simplified analytical models, where the assigned coefficients have a certain experimental support.

The objective of this work is to develop a mesoscale model (pebble scale) to study temperature transients for surveillance purposes in operation. The procedure to solve the model equations uses the trial functions with physical meaning method (Acton and Squire, 1985). The solutions are approximated, but very simple, and in general, enough for safety requirements and surveillance, which could be based on thermal noise analysis of thermocouple signals.

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At high temperatures, thermocouples such as Type C or Type S are affected by drift and this causes their lack of accuracy. Self-validation methods have been proposed to ensure metrological traceability of the measurements performed by thermocouples (Failleau et al., 2013). Nowadays, the new kind based on molybdenum and niobium (Mo/Nb) is an alternative choice due to their very low neutron absorption cross section and remain nearly unchanged by neutron flux (Villard et al., 2008). Recently, progress has been made in this regard (Sadli et al., 2013; Johansson et al., 2011), but long-term stability of these thermocouples is still insufficient in order to ensure the accuracy of the measurements. Nevertheless, for thermal noise analysis short-term stability is only required.

Once the heat transfer model is derived, the application for both thermal noise analysis and temperature transients are presented based on data from the literature (Garcia Fajardo, 2012).

2. Description of TRISO fuel

This type of fuel, known as TRISO (tristructural isotropic) coated fuel particle, is composed of micro-spheres whose diameter is 650–850 μm and made of three layers of pyrolytic carbon PyC (buffer, inner pyrolytic carbon and outer pyrolytic carbon) and one of silicon carbide (SiC) Nuclear Regulatory Commission, 2004.

The particles are inside a graphite matrix fuel element the size of a tennis ball, 6 cm diameter. The idea behind this design is that each particle is responsible for the confinement of all high burnup fission products inside this fuel (López-Honorato et al., 2009), even under accident conditions.

Above 2000 °C the probability of the failure of the coating of a TRISO pebble increases dramatically; fission product release and thermal decomposition of SiC quickly increase (Stainsby et al., 2008). This is why this temperature is taken as a reference in the accident analysis. For instance, the normal operation in the core does not exceed 1250 °C, and the temperature of helium is around 850 °C. In fact, there are several failure mechanisms that are burnup and temperature dependent such as pressure vessel failure, palladium attack, kernel migration, etc (Powers and Wirth, 2010; Maki et al., 2007).

Consequently, it is important to calculate the temperature distribution inside the pebble. The difficulty lies in knowing the coefficients of the differential equations, that is, heat capacity and thermal conductivity, since fuel deteriorates with burnup, and it is supposed that high burnup up to 5 times higher than in Light Water Reactors (LWRs) are reached. Powers establishes in Powers and Wirth (2010) that uncertainties and unknowns in material properties are abundant, especially with regard to how properties vary as functions of temperature, fast fluence and burnup.

Another aspect in the fuel different from LWRs is that TRISO particles are ceramic, so their heat conductivity is low and consequently Biot number is much higher than unity. Heat transfer times are long; in the order of 4 s in LWR, compared to 15 s in TRISO. The thermal noise is in a very low frequency range indeed.

The best calculation temperature profile procedures can fail in their forecasts due to the lack of accuracy in the coefficients. The exact calculation, as an idealized contour problem, is too complex to be practical. However, it provides hints in order to be able to use approximated problems with a certain guarantee.

3. Theoretical basis

The radial temperature profile inside the fuel ball whose radius is R is calculated formally as an eigenvalues problem. The heat transfer equation in spherical coordinates is:

$$\frac{\partial^2 T}{\partial r^2} + \frac{2}{r}\frac{\partial T}{\partial r} + \frac{q}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t} \tag{1}$$

where T is the temperature inside the fuel ball; q the thermal power density generated by the fissions; k, the heat conductivity; and α, the thermal diffusivity. Symmetry is assumed in the rest of spherical coordinates.

The equation coefficients are not always well known, especially during burnup. The goal is solving the equation in order to design an approximated solving method later, so that the conductivity is linked to the heat transfer time inside the ball. This time can be estimated on line through noise analysis techniques (Thie, 1981; IAEA, 2013; Hashemian, 2011). If there is an empirical relationship between estimated fuel burnup and the thermal conductivity, the heat transfer time measurement will provide information on the fuel burnup level.

Let T_R be the outer pebble temperature and T_f the temperature of the coolant, normally helium. The contour conditions are:

Null gradient in the ball centre: $\left(\frac{\partial T}{\partial r}\right)_{r=0} = 0$

Surface heat transfer: $k\left(\frac{\partial T}{\partial r}\right)_{r=R} = -h(T_R - T_f)$ being h the convective heat transfer coefficient between the pebble and the helium. Afterwards, the temperature is separated into the addition of two functions; the homogeneous one $H(r,t)$ and the particular one, $P(r)$, resulting:

$$T(r,t) = H(r,t) + P(r) \tag{2}$$

Taking this decomposition to the differential Eq. (1):

$$\frac{\partial^2 H}{\partial r^2} + \frac{2}{r}\frac{\partial H}{\partial r} = \frac{1}{\alpha}\frac{\partial H}{\partial t};\quad \left(\frac{\partial H}{\partial r}\right)_{r=0} = 0;$$
$$k\left(\frac{\partial H}{\partial t}\right)_{r=R} = -hH(R,t) \tag{3}$$

$$\frac{\partial^2 P}{\partial r^2} + \frac{2}{r}\frac{\partial P}{\partial r} + \frac{q}{k} = 0;\quad \left(\frac{dP}{dr}\right)_{r=0} = 0;$$
$$k\left(\frac{dP}{dr}\right)_{r=R} = -h\left[P(R) - T_f\right] \tag{4}$$

The homogeneous equation is solved conventionally through separation of variables:

$$H(r,t) = X(r)Z(t) \tag{5}$$

Defining the eigenvalue λ as:

$$\frac{1}{\alpha Z}\frac{dZ}{dt} = -\lambda^2 \tag{6}$$

$$\frac{1}{X}\left(\frac{d^2X}{dr^2} + \frac{2}{r}\frac{dX}{dr}\right) = -\lambda^2 \tag{7}$$

The following solution is obtained:

$$Z(t) = Z(0)\exp\left(-\alpha\lambda^2 t\right);\quad X(r) = X(0)\frac{\sin(\lambda r)}{\lambda r} \tag{8}$$

with the following contour condition:

$$k\left(\frac{dX}{dr}\right)_{r=R} = -hX(R) \tag{9}$$

In order to make the expressions non-dimensional, Biot number is used and it is defined as:

$$\mathrm{Bi} = \frac{hR}{k} \tag{10}$$

The eigenvalues are obtained by solving the equation:

$$\lambda R = (1 - \mathrm{Bi})\tan(\lambda R) \tag{11}$$

which has infinite number of solutions, though for our purpose it is enough to know the first eigenvalue. It must be taken into account when Bi >1 the first eigenvalue is in the $\frac{\pi}{2} < \lambda R < \pi$ region. Taking all eigenvalues, the general solution of the homogeneous part may be written as:

$$H(r,t) = \sum_{n=1}^{\infty} C_n \frac{\sin(\lambda_n r)}{\lambda_n r} \exp(-\alpha \lambda_n^2 t) \tag{12}$$

where the C_n coefficients are fitted with the initial values. It is interesting to observe that the Taylor expansion of the function $\sin(\lambda_n r)/\lambda_n r$ results in a parabolic profile for $\lambda_n r$ low values. In principle, there is more interest in such conclusions rather than solving the contour problem exactly, since there is a high uncertainty in the equation coefficients.For the particular component $P(r)$, after some cumbersome algebra, the solution is:

$$P(r) = T_f + \theta + \frac{\mathrm{Bi}}{2}\left(1 - \frac{r^2}{R^2}\right)\theta \tag{13}$$

where $\theta = qR/3h$. It can be observed that the exact solution is not useful at all, as it happens in most contour problems. For an approximated solution, the parabolic temperature profile is a good approximation.

4. Trial functions with physical meaning

The approximated solving method of contour problems with trial functions with physical meaning and fulfilling the contour conditions is well described in Acton and Squire (1985). The trial function proposed is:

$$T(r,t) = [T_R(t) - T_f(t)]f(r) + T_R(t) \tag{14}$$

where $T_R(t)$ corresponds to the outer pebble temperature, which could be measured by coupling a thermocouple; and $T_f(t)$ the temperature of helium, which can be fluctuating. The radial part is described as a parabolic profile:

$$f(r) = \frac{\mathrm{Bi}}{2}\left(1 - \frac{r^2}{R^2}\right) \tag{15}$$

This ensures that the approximate solution (14) matches the exact solution (2) in steady state. In fact, if T_R is the outer pebble temperature and T_f the helium temperature in steady state, the exact solution (2) is only the particular component (13) where the parameter θ is the difference between T_R and T_f. Therefore the profile (15) let the approximate solution (14) match the particular component (13).

Taking the approximation (14) to Eq. (1) yields:

$$\frac{dT_R}{dt} - \alpha\left[\frac{f''(r) + \frac{2}{r}f'(r)}{1+f(r)}\right]T_R = \frac{\alpha}{1+f(r)} \times \left[\frac{q}{k} + \frac{f(r)}{\alpha}\frac{dT_f}{dt} - \left(f''(r) + \frac{2}{r}f'(r)\right)T_f\right] \tag{16}$$

where f' and f'' are the first and second derivatives of f with respect to r.

In transfer function terms, the input is $T_f(t)$and the output $T_R(t)$. This function presents a real pole whose value is:

$$\tau(r) = -\frac{1+f(r)}{\alpha[f''(r) + \frac{2}{r}f'(r)]} \tag{17}$$

In the trial function method, this pole is explicitly determined by obtaining the solution for a certain r value. In order to obtain a real pole according to the first eigenvalue provided by Eq. (11), the value $r = R/\sqrt{2}$ is selected, resulting a simple and an approximated expression for the pole:

$$\tau = \frac{R^2}{3\alpha}\left(\frac{1}{\mathrm{Bi}} + \frac{1}{4}\right) \tag{18}$$

Theoretically, the pole τ can be measured through thermal noise analysis from a thermocouple since the lack of accuracy of the thermocouple due to the high temperature does not affect noise analysis performance. Then, an approximated value is inferred for the thermal diffusivity α, and from here, the thermal conductivity k, which changes with burnup, is obtained. In this way, there is a simple method to relate the conductivity with burnup and this can guarantee that the limit temperature inside the ball is not reached. Since there are several phenomena and failure mechanisms related to temperature and burnup, these could be also monitored during operation.

5. Example case

An example is studied with data from literature for this type of reactors (Garcia Fajardo, 2012). The data presented has been calculated for a Pebble Bed High Temperature Reactor (PBHTR). Firstly, the neutronic calculations and the thermal power density q are performed with MCNPX and the coolant temperature profiles through ANSYS CFX. It is also necessary to know the thermal conductivity of the fuel element (k), which is considered to be a homogeneous mixture of the different materials that compound the TRISO particles and pyrolytic graphite. Many different theories have been developed to estimate the effective thermal conductivity in mixtures of arbitrary concentration (Richter, 2003) but for the thermal analysis of sphere type fuel, the averaged volume thermal conductivity is used (Cho et al., 2009). The data used is listed in Table 1.

It can be observed that Bi = 6 reflects a low thermal conductivity in TRISO particles. This also explains the great difference between the helium temperature and the outer ball temperature. Explicitly, taking the physical parameters of Table 1 and $r = R/\sqrt{2}$ to Eq. (16), the differential equation provided by the trial functions method is:

$$\frac{dT_R}{dt} + aT_R = b\frac{dT_f}{dt} + aT_f + c \tag{19}$$

where the coefficients a, b and c are listed in Table 2.

The real pole is 0.064 s^{-1} which corresponds to a heat transfer time much higher than that of the fuel in conventional PWRs. If the temperature of helium T_f is considered to be the one in Table 1, Eq. (19) provides 1278 K for T_R in stationary conditions (a difference of 0.2% respect to T_R in Table 1).

On the other hand, the differential Eq. (19) suggests the noise analysis to be performed through an Autoregressive Moving Average model fit of order (1,1) or ARMA (1,1) model. This time series model is defined as:

$$y_i = \sum_{p=1}^{n} a_p y_{i-p} + \sum_{q=1}^{m} b_q \varepsilon_{i-q} + \varepsilon_i \tag{20}$$

where y_i is the thermocouple signal at instant i, a_p are the autoregressive (AR) coefficients, b_q are the moving average (MA) coefficients and ε_i is the driven input white noise at instant i. The parameters n and m are the orders of the AR and MA models, so in this case $m = n = 1$, that is ARMA (1,1). Since the dynamic model is already known (a transfer function with one pole and one zero), the Dynamic Data System (DDS) methodology can be applied

Table 1
Data from literature for the example case.

q	3.4×10^7 W m^{-3}
α	$8 \times \{10^{-6}$ m^2 s^{-1}
k	24.7 W m^{-1} K^{-1}
Bi	6 (dimensionless)
R	0.03 m
T_f	1209 K
T_R	1275 K

Table 2
Coefficients of the differential Eq. (19).

a	0.06 4 s^{-1}
b	0.6 (dimensionless)
c	4.40 K s^{-1}

(Montalvo et al., 2014; Wu and Ouyang, 1982). This means that in order to obtain the pole (inverse of heat transfer time), an ARMA (1,1) is assumed and the sampling time must be properly selected in order to fit the model to the signal PSD.

As a heat transfer time around 15 s is expected, a 3 s sampling time should be adequate, and rather higher than the one used in Pressurized Water Reactors (PWRs).

The thermal noise that is expected is produced by the following phenomena: the heat transfer between the coolant and the pebble depends on the convective heat transfer coefficient (h) and this also depends on the Reynolds number. As the coolant flow is very turbulent, the Reynolds number and h fluctuat. Finally, these fluctuations of h are causing the temperature fluctuations observed in the pebble. Hence, the frequency range of the temperature fluctuations would be in the thermo-hydraulic region. Some examples of thermal simulated noise in these reactors can be found in Agung (2007) and for a real reactor in Olson et al. (1982).

Besides, the heat transfer phenomena taken place is very slow, that is, it is located in a very low frequency range where the amplitude of the thermocouple signals are higher. It must be also taken into consideration the fact that thermocouples have a response time that is usually around 1 s, which should be sufficient to record the phenomena.

Another aspect to be considered is that the heat transfer time measured in situ is different from the one which is measured in laboratory, in a thermal bath, due to the Biot number.

In order to verify these conditions, thermal noise has been simulated using the following transfer function:

$$H(s) = \frac{bs + a}{s + a} \tag{20}$$

where the coefficients a and b are listed in Table 2.

The Power Spectral Density (PSD) fits perfectly an ARMA (1,1) model as it can be seen in Fig. 1.

The estimated pole is 0.064 s^{-1}, and the zero, 0.11 s^{-1}. The sampling time selected is 3 s. By choosing shorter sampling times, the poles obtained are not so satisfactory.

6. The trial function and the transients

The ceramic coating of TRISO particles begins to degrade from 2000 °C, so it is necessary to guarantee that in case of a transient, that temperature is not reached (Nuclear Regulatory Commission, 2004). Although the trial function (14) is an approximated solution of the heat transfer differential Eq. (1), it is possible to obtain with it a first approximation of the temperature distribution evolution inside the fuel, since the thermal conductivity is not known accurately.

The estimation is simple; it is enough to postulate a thermal evolution $T_f(t)$ for the coolant as a result of the transient to be studied and from here obtaining the evolution of the outer temperature $T_R(t)$ by resolving Eq. (19).

Since a and b coefficients from Eq. (19) are not known accurately, they are determined in situ through noise analysis. In order to calculate parameter c, it is also required to know the thermal conductivity (obtained through a and b) and the estimation of the thermal power q.

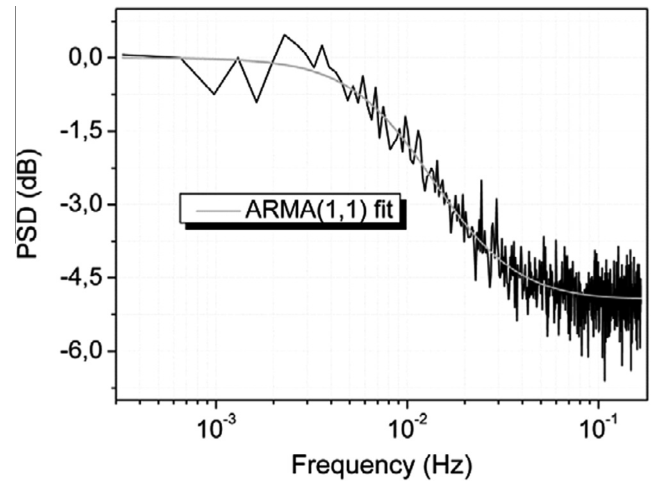

Fig. 1. PSD of simulated thermal noise fitted to an ARMA (1,1) model.

Afterwards, the trial function $T(r,t)$ to be used is the Eq. (14). In Eq. (14) it is also needed to know Bi in order to have the expression of $f(r)$. Bi is also obtained in-situ through the coefficient b.

In the case under study, the temperature of helium $T_f(t)$ is supposed to evolve as:

$$T_f(t) = T_f + \left[1 - \exp\left(-\frac{t}{\tau_f}\right)\right]\Delta T_f \tag{21}$$

where T_f is in Table 1, $\tau_f = 10$ s and $\Delta T_f = 20$ K. The initial temperature of the ball surface, T_R, is supposed to be 1278 K. The increase $T_f(t) - T_f$ can be seen in Fig. 2. The resulting increase $T_R(t) - T_R$ can also be seen in Fig. 2. $T(r,t)$ is estimated from the outer temperature at three instants: the initial one, $3\tau_f$ and $10\tau_f$(see Fig. 3).

As it can be seen, the temperature profiles calculated with the trial function method (Fig. 3 left) and with ANSYS CFX (Fig. 3 right) are very similar, in fact they are in different plots in order to properly distinguish them.

It can be noticed that the safety margin, up to 2000 °C has not been exceeded (temperature reaches 1504 K in the center of the ball at 100 s). It is possible and desirable estimating this margin with numerical methods for contour problems, which will act as a likelihood reference for the trial function.

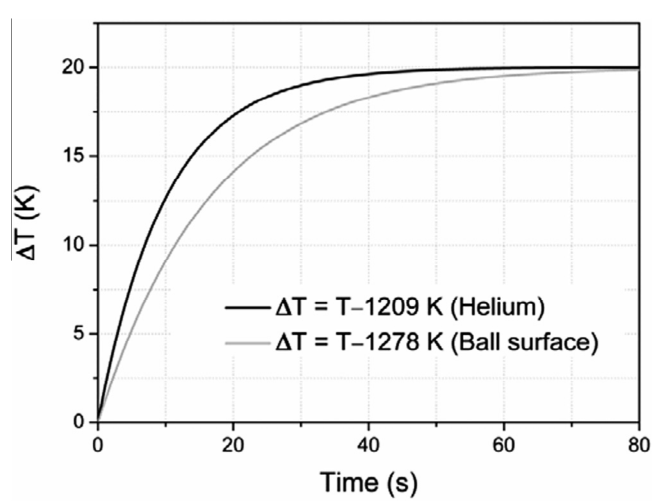

Fig. 2. Temperature increase: Helium (black) and ball surface (grey).

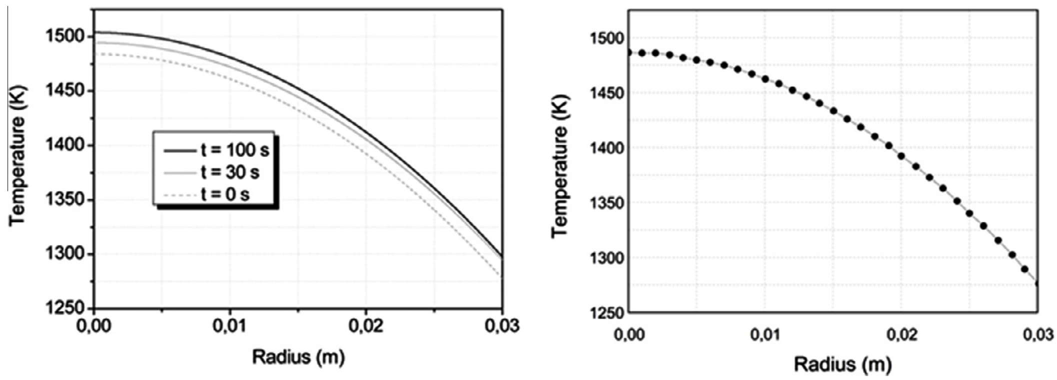

Fig. 3. Temperature distribution inside the ball at initial instant, $3\tau_f$ and $10\tau_f$ (left) and temperature distribution inside the ball calculated with ANSYS CFX (right, taken from Garcia Fajardo (2012)).

7. Conclusions

The TRISO type fuel is composed of particles with a ceramic coating which produces the heat transfer to be slow and high Biot numbers, which constrain the standard safety specifications not to exceed the coating failure temperature, in the order of 2000 °C.

A theoretical model is developed to study the temperature profiles inside a TRISO pebble. The model equations are solved approximately with the trial function method. This procedure gives a very practical solution in which the fuel and coolant temperatures are related by a transfer function with one zero and one pole. The pole is connected to the conductivity which changes with burnup. By means of thermal noise analysis from thermocouple signals, the pole and therefore, the heat transfer time can be estimated in situ. For that purpose, a time domain analysis based on an ARMA (1,1) model is chosen.

By taking data from the literature, the model proposed is simulated for both obtaining the heat transfer time and the response to certain temperature transients. The results show a heat transfer time around 15 s and the model is valid to monitor the safety limits in situ.

If the level of burnup and the conductivity is previously correlated in a laboratory, by monitoring the heat transfer time, the surveillance of the level of burnup is also accomplished.

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